

PLUS-MINUS 5V SUPPLY From 9V Battery

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Op-amps require dual-polarity supply for proper operation. When working with battery supply, it becomes

difficult to get dual power supply for the op-amps. Presented here is a simple circuit that provides $\pm 5V$ from a 9V battery.

Circuit and working

The circuit diagram for the $\pm 5V$ supply from a 9V battery is shown in Fig. 1. It is built around 9V battery (BATT.1), voltage regulator IC 78L05 (IC1), CMOS voltage converter ICL7660 (IC2) and a few other components.

Voltage regulator IC1 converts 9V battery input into regulated 5V. This 5V output from IC1 is given to pin 8 of IC2. IC2 and capacitors C3 and C4 form the voltage inverter section that converts +5V to -5V. Converted -5V supply is available at pin 5 of IC2. Converted $\pm 5V$ supply is thus available at connector CON1.

Construction and testing

An actual-size, single-side PCB for $\pm 5V$ supply from 9V battery is shown in Fig. 2 and its component layout in Fig. 3.

Assemble the circuit on the PCB and enclose it in a water-proof box. Battery BATT.1 should be enclosed in the box. Fix CON1 on the front or rear side of the cabinet, so that you can use the $\pm 5V$ easily.

Before using the circuit, verify the test points given in the table to ensure proper working of the circuit. **EFY**

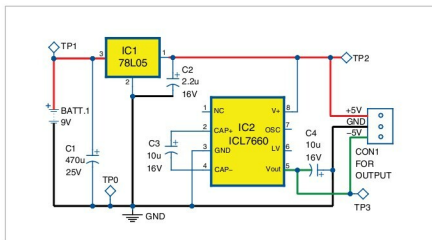


Fig. 1: Circuit diagram of $\pm 5V$ supply from 9V battery

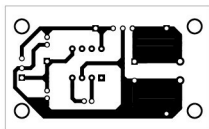


Fig. 2: Actual-size PCB layout of $\pm 5V$ supply from 9V battery

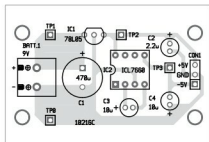


Fig. 3: Component layout of the PCB

Test Points

Test point	Details
TP0	0V (GND)
TP1	9V
TP2	+5V
TP3	-5V

PARTS LIST

Semiconductors:	
IC1	- 78L05 voltage regulator
IC2	- ICL7660 voltage converter
Capacitors:	
C1	- 470 μ F, 25V electrolytic
C2	- 2.2 μ F, 16V electrolytic
C3, C4	- 10 μ F, 16V electrolytic
Miscellaneous:	
CON1	- 3-pin connector terminal
BATT.1	- 9V battery



A. Samiuddin is B.Tech in electrical and electronics engineering. He has been an electronics hobbyist since ninth standard.